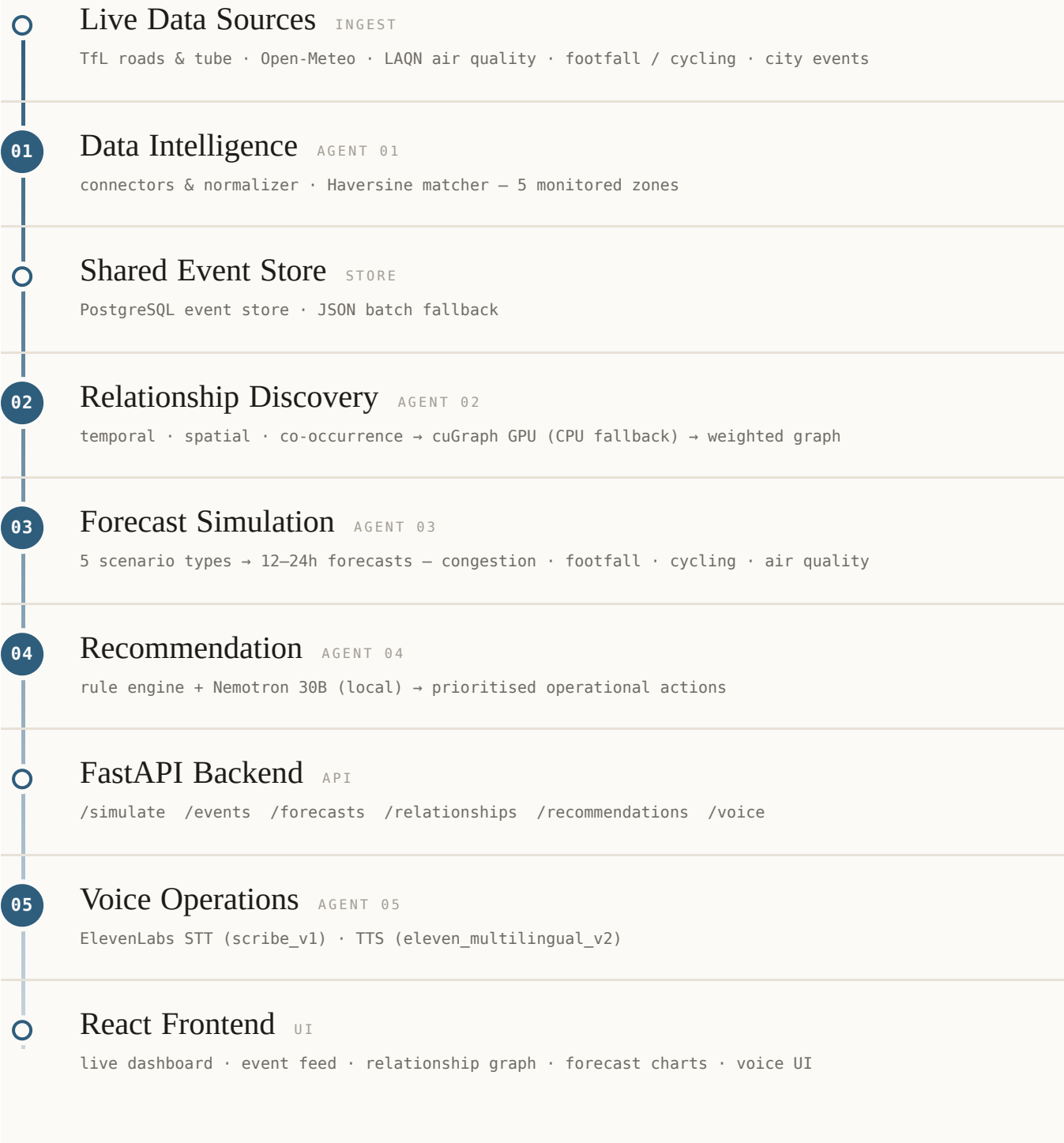


CitiSim

System Architecture

A grounded, locally-run simulation engine: live city signals flow through five cooperating agents — from ingestion and relationship discovery to forecasting, recommendation, and voice — backed by a shared event store with fallbacks at every layer.



Key Design Decisions

DECISION	REASON
Nemotron runs locally	City operational data stays on-premise — no sensitive data leaves the machine.
GPU path via cuDF / cuGraph	In-memory dataframe ops and graph centrality run on NVIDIA hardware when available; CPU fallback for portability.
Five dedicated agents on a shared event store	Clean data contracts — each agent can be developed, tested, and scaled independently.
Grounded LLM — only discovered relationships reach Nemotron	Eliminates hallucination; the model cannot invent city relationships the data didn't surface.
Graceful fallbacks at every layer	No Nemotron → rule-based · no Postgres → JSON batch · no ElevenLabs → typed prompt.